

Script-1: loop.py

```
#!/usr/bin/env python3
import subprocess, time, os, signal

QEMU_CMD = [
    "qemu-system-riscv64",
    "-machine", "virt,accel=tcg",
    "-cpu", "rv64",
    "-m", "2048",
    "-kernel", os.path.expanduser("~/kernel-build/linux-6.14.5/arch/riscv/boot/Image"),
    "-drive", f"file={os.path.expanduser('~/.ubuntu-24.04.2-preinstalled-server-riscv64+nezha.img')},f>    "-append",
    "root=/dev/vda1 console=ttyS0,115200 rw rootwait",
    "-serial", "file:/mnt/f/loggins/console_log_timi.txt",
    "-nographic",
    "-qmp", "unix:/tmp/qmp-sock,server=on,wait=off"
]

ITERATIONS = 3

for i in range(ITERATIONS):
    print(f"\n--- Starting QEMU for Iteration {i+1} ---")
    try:
        if os.path.exists("/tmp/qmp-sock"):
            os.remove("/tmp/qmp-sock")

        qemu = subprocess.Popen(QEMU_CMD)
        time.sleep(6) # Boot delay

        print(f"--- Running timi.py (Iteration {i+1}) ---")
        subprocess.run(["python3", "timi.py"], check=True)

    finally:
        print(f"--- Killing QEMU (Iteration {i+1}) ---")
        qemu.send_signal(signal.SIGTERM)
        qemu.wait()
        time.sleep(2)
```

Script:2- timi.py

```
#!/usr/bin/env python3
import time, socket, json, re, logging
from docx import Document
from datetime import datetime

QMP_SOCKET = "/tmp/qmp-sock"
REGISTERS = ["ra", "sp", "gp", "tp", "t0", "t1", "t2", "s0", "s1", "a0", "a1", "a2", "a3"]
REPORT_PATH = "/mnt/f/loggins/timing_violation_report.docx"
MAX_RETRIES = 20
SLEEP_BETWEEN = 0.5

logging.basicConfig(level=logging.INFO, format="%(asctime)s %(message)s")
print("== Fault Injection Starting ==")

def qmp_connect():
    sock = socket.socket(socket.AF_UNIX, socket.SOCK_STREAM)
    sock.connect(QMP_SOCKET)
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sock.settimeout(2)
_ = sock.recv(4096) # greeting
sock.send(json.dumps({"execute": "qmp_capabilities"}).encode() + b"\n")
_ = sock.recv(4096)
return sock

def send_qmp(sock, cmd):
    sock.send((json.dumps(cmd) + "\n").encode())
    return sock.recv(4096).decode(errors='ignore')

def get_gprs(raw_text):
    """Extract register values from xN/NAME format."""
    matches = re.findall(r"x\d+/(w+)\s+([0-9a-f]{16})", raw_text, re.IGNORECASE)
    return {name: int(val, 16) for name, val in matches if name in REGISTERS}

def flip_bit(val, bit=63):
    return val ^ (1 << bit)

sock = qmp_connect()

doc = Document()
doc.add_heading("Timing Violation Fault Injection Report", 0)
doc.add_paragraph(f"Start time: {datetime.now().isoformat()}")
doc.add_paragraph(f"Registers: {' '.join(REGISTERS)}")
doc.add_paragraph(f"Bit flipped: MSB (bit 63)")
doc.add_paragraph("")

for iteration in range(1, 4):
    logging.info(f"--- Iteration {iteration}/3 ---")
    doc.add_heading(f"Iteration {iteration}", level=1)

    send_qmp(sock, {"execute": "stop"})
    time.sleep(SLEEP_BETWEEN)

    regs = {}
    for attempt in range(MAX_RETRIES):
        out = send_qmp(sock, {
            "execute": "human-monitor-command",
            "arguments": {"command-line": "info registers"}
        })
        regs = get_gprs(out)
        if regs:
            break
        time.sleep(SLEEP_BETWEEN)

    if not regs:
        logging.warning("No valid registers parsed.")
        doc.add_paragraph("No valid registers parsed.")
    else:
        for reg in REGISTERS:
            if reg not in regs:
                continue
            val = regs[reg]
            flipped = flip_bit(val)
            cmd = f"set ${reg} = 0x{flipped:016x}"
            send_qmp(sock, {
                "execute": "human-monitor-command",
                "arguments": {"command-line": cmd}
            })

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```
}  
log = f"{reg}: 0x{val:016x} → 0x{flipped:016x}"  
logging.info(log)  
doc.add_paragraph(log)  
  
send_qmp(sock, {"execute": "cont"})  
time.sleep(SLEEP_BETWEEN)  
  
doc.save(REPORT_PATH)  
logging.info(f"Report saved to {REPORT_PATH}")  
print("== All iterations complete. Report ready ==")
```

Result of Fault Injection:

Timing Violation Fault Injection Report

Start time: 2025-06-10T15:13:41.123310

Registers: ra, sp, gp, tp, t0, t1, t2, s0, s1, a0, a1, a2, a3

Bit flipped: MSB (bit 63)

Iteration 1

ra: 0x00007fffb5286218 → 0x80007fffb5286218

sp: 0x00007fffe5e0b8f0 → 0x80007fffe5e0b8f0

gp: 0x0000555559d3d800 → 0x8000555559d3d800

tp: 0x00007fffb46ccbe0 → 0x80007fffb46ccbe0

t0: 0x00000000a00886cf → 0x80000000a00886cf

t1: 0x00007fffb517786c → 0x80007fffb517786c

t2: 0x000000002fe46a6b → 0x800000002fe46a6b

s0: 0x00007fffe5e0b960 → 0x80007fffe5e0b960

s1: 0x0000555559d413b0 → 0x8000555559d413b0

a0: 0x00007fffb5596660 → 0x80007fffb5596660

a1: 0x0000000000000010 → 0x8000000000000010

a2: 0x00007fffe5e0b8f8 → 0x80007fffe5e0b8f8

a3: 0x0000000000000000 → 0x8000000000000000

Iteration 2

ra: 0xffffffff805da4d2 → 0x7fffffff805da4d2

sp: 0xff20000000003d60 → 0x7f20000000003d60

gp: 0xffffffff81512900 → 0x7fffffff81512900

tp: 0xff600000018b8000 → 0x7f600000018b8000

t0: 0x0000000000000040 → 0x8000000000000040

t1: 0x0000000000000000 → 0x8000000000000000

t2: 0xffffffff80e013a0 → 0x7fffffff80e013a0

s0: 0xff20000000003d70 → 0x7f20000000003d70

s1: 0xffffffff8157bca0 → 0x7fffffff8157bca0

a0: 0xffffffff8157bca0 → 0x7fffffff8157bca0

a1: 0x0000000000000000 → 0x8000000000000000

a2: 0x0000000000000006e → 0x8000000000000006e

a3: 0xff600000024b4000 → 0x7f600000024b4000

Iteration 3

ra: 0xffffffff8019d16e → 0x7fffffff8019d16e

sp: 0xff20000000353e20 → 0x7f20000000353e20

gp: 0xffffffff81512900 → 0x7fffffff81512900

tp: 0xff60000002700000 → 0x7f60000002700000

t0: 0x0000000000000000 → 0x8000000000000000

t1: 0x000000000000000a → 0x800000000000000a

t2: 0x0000000000000000 → 0x8000000000000000

s0: 0xff20000000353e60 → 0x7f20000000353e60

s1: 0x00000000000001255 → 0x80000000000001255

a0: 0xff60000002700000 → 0x7f60000002700000

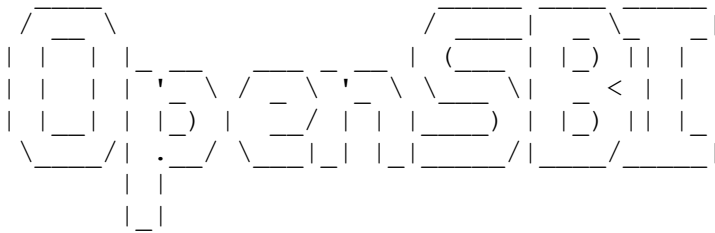
a1: 0x0000000000000000 → 0x8000000000000000

a2: 0xffffffffffffff → 0x7fffffffffffff

a3: 0x00000000000003383 → 0x80000000000003383

Console Log:

OpenSBI v1.5.1



```
Platform Name           : riscv-virtio,qemu
Platform Features       : medeleg
Platform HART Count     : 1
Platform IPI Device     : aclint-mswi
Platform Timer Device   : aclint-mtimer @ 10000000Hz
Platform Console Device : uart8250
Platform HSM Device     : ---
Platform PMU Device     : ---
Platform Reboot Device  : syscon-reboot
Platform Shutdown Device: syscon-poweroff
Platform Suspend Device : ---
Platform CPPC Device    : ---
Firmware Base          : 0x80000000
Firmware Size          : 327 KB
Firmware RW Offset     : 0x40000
Firmware RW Size       : 71 KB
Firmware Heap Offset   : 0x49000
Firmware Heap Size     : 35 KB (total), 2 KB (reserved), 11 KB (used), 21 KB
                        (free)
Firmware Scratch Size  : 4096 B (total), 416 B (used), 3680 B (free)
Runtime SBI Version    : 2.0

Domain0 Name           : root
Domain0 Boot HART      : 0
Domain0 HARTs          : 0*
Domain0 Region00       : 0x0000000000100000-0x0000000000100fff M: (I,R,W) S/U:
                        (R,W)
Domain0 Region01       : 0x0000000001000000-0x0000000001000fff M: (I,R,W) S/U:
                        (R,W)
Domain0 Region02       : 0x0000000002000000-0x000000000200ffff M: (I,R,W) S/U:
                        ()
Domain0 Region03       : 0x00000000080040000-0x0000000008005ffff M: (R,W) S/U: ()
Domain0 Region04       : 0x00000000080000000-0x0000000008003ffff M: (R,X) S/U: ()
Domain0 Region05       : 0x000000000c400000-0x000000000c5fffff M: (I,R,W) S/U:
                        (R,W)
Domain0 Region06       : 0x000000000c000000-0x000000000c3fffff M: (I,R,W) S/U:
                        (R,W)
Domain0 Region07       : 0x0000000000000000-0xffffffffffffffff M: () S/U:
                        (R,W,X)
Domain0 Next Address    : 0x00000000080200000
Domain0 Next Arg1      : 0x00000000ffe00000
Domain0 Next Mode       : S-mode
Domain0 SysReset        : yes
Domain0 SysSuspend     : yes

Boot HART ID           : 0
Boot HART Domain       : root
Boot HART Priv Version  : v1.12
Boot HART Base ISA     : rv64imafdc
Boot HART ISA Extensions : sstc,zicntr,zihpm,zicboz,zicbom,sdtrig,svadu
Boot HART PMP Count    : 16
Boot HART PMP Granularity : 2 bits
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Boot HART PMP Address Bits: 54
Boot HART MHPM Info      : 16 (0x0007fff8)
Boot HART Debug Triggers : 2 triggers
Boot HART MIDELEG        : 0x00000000000001666
Boot HART MEDELEG        : 0x00000000000f0b509
[ 0.000000] Linux version 6.14.5 (kazumin@PR3D4T0R) (riscv64-linux-gnu-gcc
(Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #1
SMP Fri May  2 19:37:40 CEST 2025
[ 0.000000] random: crng init done
[ 0.000000] Machine model: riscv-virtio,gemu
[ 0.000000] SBI specification v2.0 detected
[ 0.000000] SBI implementation ID=0x1 Version=0x10005
[ 0.000000] SBI TIME extension detected
[ 0.000000] SBI IPI extension detected
[ 0.000000] SBI RFENCE extension detected
[ 0.000000] SBI SRST extension detected
[ 0.000000] SBI DBCN extension detected
[ 0.000000] efi: UEFI not found.
[ 0.000000] OF: reserved mem: 0x0000000080000000..0x000000008003ffff (256 KiB)
nomap non-reusable mmode_resv1@80000000
[ 0.000000] OF: reserved mem: 0x0000000080040000..0x000000008005ffff (128 KiB)
nomap non-reusable mmode_resv0@80040000
[ 0.000000] Zone ranges:
[ 0.000000]   DMA32      [mem 0x0000000080000000-0x00000000ffffffff]
[ 0.000000]   Normal    empty
[ 0.000000] Movable zone start for each node
[ 0.000000] Early memory node ranges
[ 0.000000]   node 0: [mem 0x0000000080000000-0x000000008005ffff]
[ 0.000000]   node 0: [mem 0x0000000080060000-0x00000000ffffffff]
[ 0.000000] Initmem setup node 0 [mem 0x0000000080000000-0x00000000ffffffff]
[ 0.000000] SBI HSM extension detected
[ 0.000000] riscv: base ISA extensions acdfhim
[ 0.000000] riscv: ELF capabilities acdfim
[ 0.000000] Queued spinlock using Ziccrse: enabled
[ 0.000000] percpu: Embedded 22 pages/cpu s49448 r8192 d32472 u90112
[ 0.000000] Kernel command line: root=/dev/vda1 console=ttyS0,115200 rw rootwait
[ 0.000000] printk: log buffer data + meta data: 131072 + 458752 = 589824 bytes
[ 0.000000] Dentry cache hash table entries: 262144 (order: 9, 2097152 bytes,
linear)
[ 0.000000] Inode-cache hash table entries: 131072 (order: 8, 1048576 bytes,
linear)
[ 0.000000] Built 1 zonelists, mobility grouping on. Total pages: 524288
[ 0.000000] mem auto-init: stack:all(zero), heap alloc:off, heap free:off
[ 0.000000] software IO TLB: SWIOTLB bounce buffer size adjusted to 2MB
[ 0.000000] software IO TLB: area num 1.
[ 0.000000] software IO TLB: mapped [mem 0x00000000fca00000-0x00000000fcc00000]
(2MB)
[ 0.000000] Virtual kernel memory layout:
[ 0.000000]   fixmap : 0xff1bfffffea00000 - 0xff1bffffffffff000000 (6144 kB)
[ 0.000000]   pci io  : 0xff1bffffffffff000000 - 0xff1c00000000000000 ( 16 MB)
[ 0.000000]   vmemmap : 0xff1c00000000000000 - 0xff2000000000000000 (1024 TB)
[ 0.000000]   vmlalloc : 0xff2000000000000000 - 0xff6000000000000000 (16384 TB)
[ 0.000000]   modules : 0xfffffffff0158f000 - 0xfffffffff800000000 (2026 MB)
[ 0.000000]   lowmem  : 0xff6000000000000000 - 0xff6000008000000000 (2048 MB)
[ 0.000000]   kernel  : 0xfffffffff800000000 - 0xfffffffffffffffffffff (2047 MB)
[ 0.000000] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=1, Nodes=1
[ 0.000000] rcu: Hierarchical RCU implementation.
[ 0.000000] rcu: RCU restricting CPUs from NR_CPUS=64 to nr_cpu_ids=1.
[ 0.000000] rcu: RCU debug extended QS entry/exit.
[ 0.000000] Tracing variant of Tasks RCU enabled.
[ 0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 25
jiffies.
[ 0.000000] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=1
[ 0.000000] RCU Tasks Trace: Setting shift to 0 and lim to 1
rcu_task_cb_adjust=1 rcu_task_cpu_ids=1.
[ 0.000000] NR_IRQS: 64, nr_irqs: 64, preallocated irqs: 0

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[ 0.000000] riscv-intc: 64 local interrupts mapped
[ 0.000000] riscv: providing IPIs using SBI IPI extension
[ 0.000000] rcu: srcu_init: Setting srcu_struct sizes based on contention.
[ 0.000000] clocksource: riscv_clocksource: mask: 0xffffffffffffffff max_cycles:
0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000081] sched_clock: 64 bits at 10MHz, resolution 100ns, wraps every
4398046511100ns
[ 0.000181] riscv-timer: Timer interrupt in S-mode is available via sstc
extension
[ 0.005569] Console: colour dummy device 80x25
[ 0.009575] Calibrating delay loop (skipped), value calculated using timer
frequency.. 20.00 BogoMIPS (lpj=40000)
[ 0.009694] pid_max: default: 32768 minimum: 301
[ 0.011015] LSM: initializing lsm=capability
[ 0.013568] Mount-cache hash table entries: 4096 (order: 3, 32768 bytes, linear)
[ 0.013610] Mountpoint-cache hash table entries: 4096 (order: 3, 32768 bytes,
linear)
[ 0.051180] riscv: ELF compat mode supported
[ 0.051390] ASID allocator using 16 bits (65536 entries)
[ 0.052764] rcu: Hierarchical SRCU implementation.
[ 0.052791] rcu: Max phase no-delay instances is 1000.
[ 0.056115] EFI services will not be available.
[ 0.058337] smp: Bringing up secondary CPUs ...
[ 0.058856] smp: Brought up 1 node, 1 CPU
[ 0.062622] Memory: 2016320K/2097152K available (10189K kernel code, 4941K
rwdata, 4096K rodata, 2307K init, 484K bss, 77384K reserved, 0K cma-reserved)
[ 0.071211] devtmpfs: initialized
[ 0.081513] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff,
max_idle_ns: 7645041785100000 ns
[ 0.081614] futex hash table entries: 256 (order: 2, 16384 bytes, linear)
[ 0.084403] pinctrl core: initialized pinctrl subsystem
[ 0.091153] DMI not present or invalid.
[ 0.092816] NET: Registered PF_NETLINK/PF_ROUTE protocol family
[ 0.095988] DMA: preallocated 256 KiB GFP_KERNEL pool for atomic allocations
[ 0.096264] DMA: preallocated 256 KiB GFP_KERNEL|GFP_DMA32 pool for atomic
allocations
[ 0.096479] audit: initializing netlink subsys (disabled)
[ 0.099879] thermal_sys: Registered thermal governor 'step_wise'
[ 0.100658] audit: type=2000 audit(0.080:1): state=initialized audit_enabled=0
res=1
[ 0.101258] cpuidle: using governor menu
[ 0.128203] cpu0: Ratio of byte access time to unaligned word access is 7.05,
unaligned accesses are fast
[ 0.148457] HugeTLB: registered 2.00 MiB page size, pre-allocated 0 pages
[ 0.148484] HugeTLB: 28 KiB vmemmap can be freed for a 2.00 MiB page
[ 0.152290] ACPI: Interpreter disabled.
[ 0.152710] iommu: Default domain type: Translated
[ 0.152738] iommu: DMA domain TLB invalidation policy: strict mode
[ 0.154880] SCSI subsystem initialized
[ 0.156331] usbcore: registered new interface driver usbfs
[ 0.156569] usbcore: registered new interface driver hub
[ 0.156693] usbcore: registered new device driver usb
[ 0.158654] Advanced Linux Sound Architecture Driver Initialized.
[ 0.170071] vgaarb: loaded
[ 0.173150] clocksource: Switched to clocksource riscv_clocksource
[ 0.175503] pnp: PnP ACPI: disabled
[ 0.195593] NET: Registered PF_INET protocol family
[ 0.197204] IP idents hash table entries: 32768 (order: 6, 262144 bytes, linear)
[ 0.203436] tcp_listen_portaddr_hash hash table entries: 1024 (order: 3, 32768
bytes, linear)
[ 0.203557] Table-perturb hash table entries: 65536 (order: 6, 262144 bytes,
linear)
[ 0.203629] TCP established hash table entries: 16384 (order: 5, 131072 bytes,
linear)
[ 0.203830] TCP bind hash table entries: 16384 (order: 8, 1048576 bytes, linear)
[ 0.205161] TCP: Hash tables configured (established 16384 bind 16384)

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[ 0.205974] UDP hash table entries: 1024 (order: 5, 163840 bytes, linear)
[ 0.206375] UDP-Lite hash table entries: 1024 (order: 5, 163840 bytes, linear)
[ 0.208240] NET: Registered PF_UNIX/PF_LOCAL protocol family
[ 0.211025] RPC: Registered named UNIX socket transport module.
[ 0.211067] RPC: Registered udp transport module.
[ 0.211076] RPC: Registered tcp transport module.
[ 0.211084] RPC: Registered tcp-with-tls transport module.
[ 0.211091] RPC: Registered tcp NFSv4.1 backchannel transport module.
[ 0.211217] PCI: CLS 0 bytes, default 64
[ 0.218272] workingset: timestamp_bits=46 max_order=19 bucket_order=0
[ 0.222864] NFS: Registering the id_resolver key type
[ 0.223730] Key type id_resolver registered
[ 0.223761] Key type id_legacy registered
[ 0.224036] nfs4filelayout_init: NFSv4 File Layout Driver Registering...
[ 0.224225] nfs4flexfilelayout_init: NFSv4 Flexfile Layout Driver Registering...
[ 0.224757] 9p: Installing v9fs 9p2000 file system support
[ 0.226251] NET: Registered PF_ALG protocol family
[ 0.226520] Block layer SCSI generic (bsg) driver version 0.4 loaded (major 245)
[ 0.227005] io scheduler mq-deadline registered
[ 0.227078] io scheduler kyber registered
[ 0.227294] io scheduler bfq registered
[ 0.230542] riscv-plic: plic@c000000: mapped 95 interrupts with 1 handlers for 2
contexts.
[ 0.233668] pci-host-generic 30000000.pci: host bridge /soc/pci@30000000 ranges:
[ 0.234331] pci-host-generic 30000000.pci:      IO 0x0003000000..0x000300ffff -
> 0x000000000000
[ 0.234734] pci-host-generic 30000000.pci:      MEM 0x0040000000..0x007fffffffff -
> 0x004000000000
[ 0.234821] pci-host-generic 30000000.pci:      MEM 0x0400000000..0x07ffffffffff -
> 0x040000000000
[ 0.235263] pci-host-generic 30000000.pci: Memory resource size exceeds max for
32 bits
[ 0.235620] pci-host-generic 30000000.pci: ECAM at [mem 0x30000000-0x3ffffffff]
for [bus 00-ff]
[ 0.237059] pci-host-generic 30000000.pci: PCI host bridge to bus 0000:00
[ 0.237335] pci_bus 0000:00: root bus resource [bus 00-ff]
[ 0.237404] pci_bus 0000:00: root bus resource [io 0x0000-0xffff]
[ 0.237441] pci_bus 0000:00: root bus resource [mem 0x40000000-0x7fffffffff]
[ 0.237451] pci_bus 0000:00: root bus resource [mem 0x400000000-0x7ffffffffff]
[ 0.238694] pci 0000:00:00.0: [1b36:0008] type 00 class 0x060000 conventional
PCI endpoint
[ 0.241893] pci 0000:00:01.0: [1af4:1001] type 00 class 0x010000 conventional
PCI endpoint
[ 0.242108] pci 0000:00:01.0: BAR 0 [io 0x0000-0x007f]
[ 0.242147] pci 0000:00:01.0: BAR 1 [mem 0x00000000-0x00000fff]
[ 0.242182] pci 0000:00:01.0: BAR 4 [mem 0x00000000-0x00003fff 64bit pref]
[ 0.244021] pci 0000:00:01.0: BAR 4 [mem 0x400000000-0x400003fff 64bit pref]:
assigned
[ 0.244293] pci 0000:00:01.0: BAR 1 [mem 0x40000000-0x40000fff]: assigned
[ 0.244327] pci 0000:00:01.0: BAR 0 [io 0x0080-0x00ff]: assigned
[ 0.244590] pci_bus 0000:00: resource 4 [io 0x0000-0xffff]
[ 0.244620] pci_bus 0000:00: resource 5 [mem 0x40000000-0x7fffffffff]
[ 0.244631] pci_bus 0000:00: resource 6 [mem 0x400000000-0x7ffffffffff]
[ 0.249524] SBI CPPC extension NOT detected!!
[ 0.257268] virtio-pci 0000:00:01.0: enabling device (0000 -> 0003)
[ 0.338549] Serial: 8250/16550 driver, 4 ports, IRQ sharing disabled
[ 0.348743] printk: legacy console [ttyS0] disabled
[ 0.351600] 10000000.serial: ttyS0 at MMIO 0x10000000 (irq = 13, base_baud =
230400) is a 16550A
[ 0.353213] printk: legacy console [ttyS0] enabled
[ 1.794054] loop: module loaded
[ 1.799418] virtio_blk virtio0: 1/0/0 default/read/poll queues
[ 1.812064] virtio_blk virtio0: [vda] 9437184 512-byte logical blocks (4.83
GB/4.50 GiB)
[ 1.842754] vda: vda1 vda12 vda13 vda14 vda15 vda16
[ 1.859728] e1000e: Intel(R) PRO/1000 Network Driver

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[ 1.867512] e1000e: Copyright(c) 1999 - 2015 Intel Corporation.
[ 1.878238] usbcore: registered new interface driver uas
[ 1.886753] usbcore: registered new interface driver usb-storage
[ 1.896703] mousedev: PS/2 mouse device common for all mice
[ 1.907546] goldfish_rtc 101000.rtc: registered as rtc0
[ 1.915920] goldfish_rtc 101000.rtc: setting system clock to 2025-06-09T19:13:37
UTC (1749496417)
[ 1.932709] sdhci: Secure Digital Host Controller Interface driver
[ 1.941990] sdhci: Copyright(c) Pierre Ossman
[ 1.948789] Synopsys Designware Multimedia Card Interface Driver
[ 1.958029] sdhci-pltfm: SDHCI platform and OF driver helper
[ 1.967421] usbcore: registered new interface driver usbhid
[ 1.975947] usbhid: USB HID core driver
[ 1.981908] riscv-pmu-sbi: SBI PMU extension is available
[ 1.990753] riscv-pmu-sbi: 16 firmware and 18 hardware counters
[ 1.999808] riscv-pmu-sbi: Perf sampling/filtering is not supported as sscof
extension is not available
[ 2.019312] NET: Registered PF_INET6 protocol family
[ 2.032979] Segment Routing with IPv6
[ 2.038668] In-situ OAM (IOAM) with IPv6
[ 2.045003] sit: IPv6, IPv4 and MPLS over IPv4 tunneling driver
[ 2.056671] NET: Registered PF_PACKET protocol family
[ 2.065094] 9pnet: Installing 9P2000 support
[ 2.071794] Key type dns_resolver registered
[ 2.114592] debug_vm_pgtable: [debug_vm_pgtable]: Validating
architecture page table helpers
[ 2.138962] clk: Disabling unused clocks
[ 2.145186] PM: genpd: Disabling unused power domains
[ 2.152556] ALSA device list:
[ 2.156931]   No soundcards found.
[ 2.195220] EXT4-fs (vda1): recovery complete
[ 2.202877] EXT4-fs (vda1): mounted filesystem 626ef4b6-84d2-4b70-bd20-
e9f892339465 r/w with ordered data mode. Quota mode: disabled.
[ 2.221422] VFS: Mounted root (ext4 filesystem) on device 254:1.
[ 2.233393] devtmpfs: mounted
[ 2.280062] Freeing unused kernel image (initmem) memory: 2304K
[ 2.289729] Run /sbin/init as init process
[ 2.541692] systemd[1]: systemd 255.4-lubuntu8.5 running in system mode (+PAM
+AUDIT +SELINUX +APPARMOR +IMA +SMACK +SECCOMP +GCRYPT -GNUTLS +OPENSSL +ACL +BLKID
+CURL +ELFUTILS +FIDO2 +IDN2 -IDN +IPTC +KMOD +LIBCRYPTSETUP +LIBFDISK +PCRE2 -
PWQUALITY +P11KIT +QRENCODE +TPM2 +BZIP2 +LZ4 +XZ +ZLIB +ZSTD -BPF_FRAMEWORK -
XKBCOMMON +UTMP +SYSVINIT default-hierarchy=unified)
[ 2.587426] systemd[1]: Detected virtualization qemu.
[ 2.595593] systemd[1]: Detected architecture riscv64.

```

Welcome to [lmUbuntu 24.04.2 LTS[0m!

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[ 2.617978] systemd[1]: Hostname set to <ubuntu>.
[ 4.885989] systemd[1]: Queued start job for default target graphical.target.
[ 4.952404] systemd[1]: Created slice system-modprobe.slice - Slice
/system/modprobe.
[[0;32m OK [0m] Created slice [0;1;39msystem-modprobe.slice[0m - Slice
/system/modprobe.

[ 4.977948] systemd[1]: Created slice system-serial\x2dgetty.slice - Slice
/system/serial-getty.
[[0;32m OK [0m] Created slice [0;1;39msystem-serial\x2dgetty.slice[0m - Slice
/system/serial-getty.

[ 5.003350] systemd[1]: Created slice system-systemd\x2dfsck.slice - Slice
/system/systemd-fsck.
[[0;32m OK [0m] Created slice [0;1;39msystem-systemd\x2dfsck.slice[0m - Slice
/system/systemd-fsck.

[ 5.027449] systemd[1]: Created slice user.slice - User and Session Slice.

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[0;32m OK [0m Created slice [0;1;39muser.slice[0m - User and Session Slice.

[ 5.046595] systemd[1]: Started systemd-ask-password-wall.path - Forward
Password Requests to Wall Directory Watch.
[[0;32m OK [0m Started [0;1;39msystemd-ask-password-wall.â€¦[0md Requests to
Wall Directory Watch.

[ 5.071983] systemd[1]: proc-sys-fs-binfmt_misc.automount - Arbitrary Executable
File Formats File System Automount Point was skipped because of an unmet condition
check (ConditionPathExists=/proc/sys/fs/binfmt_misc).
[ 5.097644] systemd[1]: Expecting device dev-disk-by\x2dlabel-UEFI.device -
/dev/disk/by-label/UEFI...
    Expecting device [0;1;39mdev-disk-by\x2dlaâ€¦device[0m - /dev/disk/by-
label/UEFI...

[ 5.124424] systemd[1]: Expecting device dev-ttyS0.device - /dev/ttyS0...
    Expecting device [0;1;39mdev-ttyS0.device[0m - /dev/ttyS0...

[ 5.144601] systemd[1]: Reached target integritysetup.target - Local Integrity
Protected Volumes.
[[0;32m OK [0m Reached target [0;1;39mintegritysetup.targâ€¦[0m Local Integrity
Protected Volumes.

[ 5.172733] systemd[1]: Reached target slices.target - Slice Units.
[[0;32m OK [0m Reached target [0;1;39mslices.target[0m - Slice Units.

[ 5.192542] systemd[1]: Reached target snapd.mounts-pre.target - Mounting snaps.
[[0;32m OK [0m Reached target [0;1;39msnapd.mounts-pre.target[0m - Mounting
snaps.

[ 5.215409] systemd[1]: Reached target snapd.mounts.target - Mounted snaps.
[[0;32m OK [0m Reached target [0;1;39msnapd.mounts.target[0m - Mounted snaps.

[ 5.237218] systemd[1]: Reached target swap.target - Swaps.
[[0;32m OK [0m Reached target [0;1;39mswap.target[0m - Swaps.

[ 5.260340] systemd[1]: Reached target veritysetup.target - Local Verity
Protected Volumes.
[[0;32m OK [0m Reached target [0;1;39mveritysetup.target[0m - Local Verity
Protected Volumes.

[ 5.288727] systemd[1]: Listening on dm-event.socket - Device-mapper event
daemon FIFOs.
[[0;32m OK [0m Listening on [0;1;39mdm-event.socket[0m - Device-mapper event
daemon FIFOs.

[ 5.315974] systemd[1]: Listening on lvm2-lvmpolld.socket - LVM2 poll daemon
socket.
[[0;32m OK [0m Listening on [0;1;39mlvm2-lvmpolld.socket[0m - LVM2 poll daemon
socket.

[ 5.343238] systemd[1]: Listening on multipathd.socket - multipathd control
socket.
[[0;32m OK [0m Listening on [0;1;39mmultipathd.socket[0m - multipathd control
socket.

[ 5.369797] systemd[1]: Listening on syslog.socket - Syslog Socket.
[[0;32m OK [0m Listening on [0;1;39msyslog.socket[0m - Syslog Socket.

[ 5.390945] systemd[1]: Listening on systemd-fsckd.socket - fsck to fsckd
communication Socket.
[[0;32m OK [0m Listening on [0;1;39msystemd-fsckd.socket[0m â€¦fsck to fsckd
communication Socket.

[ 5.419357] systemd[1]: Listening on systemd-initctl.socket - initctl
Compatibility Named Pipe.
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[0;32m OK [0m] Listening on [0;1;39msystemd-initctl.socket - initctl Compatibility Named Pipe.

[5.447681] systemd[1]: Listening on systemd-journald-dev-log.socket - Journal Socket (/dev/log).

[0;32m OK [0m] Listening on [0;1;39msystemd-journald-dev-log.socket - Journal Socket (/dev/log).

[5.477100] systemd[1]: Listening on systemd-journald.socket - Journal Socket.

[0;32m OK [0m] Listening on [0;1;39msystemd-journald.socket - Journal Socket.

[5.502682] systemd[1]: Listening on systemd-networkd.socket - Network Service Netlink Socket.

[0;32m OK [0m] Listening on [0;1;39msystemd-networkd.socket - Network Service Netlink Socket.

[5.532146] systemd[1]: systemd-pcrextend.socket - TPM2 PCR Extension (Varlink) was skipped because of an unmet condition check (ConditionSecurity=measured-uki).

[5.556342] systemd[1]: Listening on systemd-udevd-control.socket - udev Control Socket.

[0;32m OK [0m] Listening on [0;1;39msystemd-udevd-control.socket - udev Control Socket.

[5.584352] systemd[1]: Listening on systemd-udevd-kernel.socket - udev Kernel Socket.

[0;32m OK [0m] Listening on [0;1;39msystemd-udevd-kernel.socket - udev Kernel Socket.

[5.638159] systemd[1]: Mounting dev-hugepages.mount - Huge Pages File System... Mounting [0;1;39mdev-hugepages.mount - Huge Pages File System...

[5.705246] systemd[1]: Mounting dev-mqueue.mount - POSIX Message Queue File System...

Mounting [0;1;39mdev-mqueue.mount - POSIX Message Queue File System...

[5.808947] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File System...

Mounting [0;1;39msys-kernel-debug.mount - Kernel Debug File System...

[5.861389] systemd[1]: sys-kernel-tracing.mount - Kernel Trace File System was skipped because of an unmet condition check (ConditionPathExists=/sys/kernel/tracing).

[5.947080] systemd[1]: Starting systemd-journald.service - Journal Service... Starting [0;1;39msystemd-journald.service - Journal Service...

[6.029472] systemd[1]: Starting keyboard-setup.service - Set the console keyboard layout...

Starting [0;1;39mkeyboard-setup.service - Set the console keyboard layout...

[6.108895] systemd[1]: kmod-static-nodes.service - Create List of Static Device Nodes was skipped because of an unmet condition check (ConditionFileNotEmpty=/lib/modules/6.14.5/modules.devname).

[6.221278] systemd[1]: Starting lvm2-monitor.service - Monitoring of LVM2 mirrors, snapshots etc. using dmeventd or progress polling...

Starting [0;1;39mlvm2-monitor.service - Monitoring dmeventd or progress polling...

[6.402171] systemd[1]: Starting modprobe@configfs.service - Load Kernel Module configfs...

[6.416737] systemd-journald[89]: Collecting audit messages is disabled. Starting [0;1;39mmodprobe@configfs.service - Load Kernel Module configfs...

[6.617905] systemd[1]: Starting modprobe@dm_mod.service - Load Kernel Module dm_mod...

Starting [0;1;39mmmodprobe@dm_mod.service[0m - Load Kernel Module dm_mod...

[6.825238] systemd[1]: Starting modprobe@drm.service - Load Kernel Module
drm...

Starting [0;1;39mmmodprobe@drm.service[0m - Load Kernel Module drm...

[7.025780] systemd[1]: Starting modprobe@efi_pstore.service - Load Kernel
Module efi_pstore...